

MOHAMMAD R. AL-EIADEH

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SUMMARY

Mohammad Riyad Al-Eideh is a dedicated Ph.D. candidate in the Elmore family school of Electrical and Computer Engineering at Purdue University. He earned his B.Sc. and M.Sc. in Computer Science from Al-Balqa University in 2018 and from Yarmouk University in 2021. His **research** focuses on **Graph Machine Learning, Graph Computations, Matrix Perturbation Theory, Dynamic Networks, Tensor Product of Graphs, Higher-Order Graphs, and Cybersecurity (Interdependent Systems modeled as Attack Graphs)**. He has experience on optimization problems such as Graph Embedding. He is interested in learning on graphs, Java, multi-threading, functional programming, and data structures, and is passionate about solving complex challenges in computer science. Contact him at maleiade@purdue.edu.

EDUCATION

8/2023 - present	Purdue University, Indiana, USA Ph.D. in Electrical and Computer Engineering, 90 credit hours Thesis Fields: Cybersecurity, Graph Algorithms, Attack Graphs, Resource Allocation, Machine Learning	School
9/2019 - 7/2022	Yarmouk University, Irbid, Jordan Masters' in Computer Science, 32 Credit Hours Thesis Fields: Feature Selection, Evolutionary Algorithm, Machine Learning	School
9/2015 - 7/2018	Al-Huson University College Al-Balqa Applied University, Irbid, Jordan Bachelors' in Computer Science, 132 Credit Hours Project Fields: Android, Arduino, Smart Home	School

EXPERIENCE

8/2023 - Present	Graduate Researcher & Software Developer • Securing Interdependent Systems: Developing graph-theoretic and graph-learning-based models for protecting interdependent systems represented as attack graphs.	Advisor: Dr. Mustafa Abdallah
1/2026 - Present	Graduate Teaching Assistant • Assisting in Communication Network Design & Advanced Network Security; grade homework and laboratory assignments, Provide academic support and hold office hours.	Courses: CIT 44000, CIT 40600 — Purdue University
8/2025 - 12/2025	Graduate Teaching Assistant • Assisted in Java programming, mentoring students in OOP, basic GUI development, and problem solving; evaluated coursework and provided technical support during lab sessions.	Course: CNIT 25501 — Purdue University
1/2025 - 5/2025	Graduate Teaching Assistant • Mentored students in programming laboratories, assisting with data structures, algorithms, and debugging, and providing technical guidance during lab sessions.	Purdue University
8/2024 - 12/2024	Graduate Teaching Assistant • Supported instruction in Python and MATLAB for first-year engineering students, helping develop computational thinking and engineering problem-solving skills.	Course: ENGR133 — Purdue University
8/2022 - 8/2023	Researcher & Software Developer • Developed hybrid evolutionary optimization algorithms for NP-hard combinatorial problems, including Maximum Clique (DIMACS), Feature Selection & 0-1 Knapsack.	Advisor: Prof. Mohammad Batah

AWARDS & HONORS

- 7/2023 **Purdue School of Engineering and Technology, IUPUI, IN, USA**
Graduate Research Assistant-ship, PhD in ECE at E & T Department
- 6/2018 **Al-Huson University College, Al-Huson, Jordan**
Academic Excellence Award - Ranked 1st in my class for academic performance

PHD COURSES

ECE 62900 – INTRO NEURAL NETWORKS
ECE 66200 – PATTN RECOG / DECIS PROC
MA 51100 – LINEAR ALGEBRA APPL
ECE 58000 – OPT METH FOR SYS & CON
ECE 59500 – INTRODUCTION TO GAME THEORY
MA 57500 – GRAPH THEORY

PUBLICATIONS

- Journal Article **Mohammad Ryiad Al-Eiadeh** and Mustafa Abdallah. "Auto-Sec-HS: Meta-Learning Approach Based on Hand-Crafted Domain-Based Features for Efficient Security Resource Allocation on Attack Graphs," *Journal of Information Security and Applications (JISA)*, **Under Review**, 2026.
- Journal Article **Mohammad Ryiad Al-Eiadeh** and Mustafa Abdallah. "A Robustness Analysis of Attack-Graph-Driven Security Resource Allocation Schemes under Adversarial Attacks," *ACM Transactions on Privacy and Security (TOPS)*, **Under Review**, 2026
- Journal Article **Mohammad Ryiad Al-Eiadeh** and Mustafa Abdallah. Auto-Sec: Meta-Learning for Automated Efficient Security Resource Allocation on Attack Graphs. *IEEE Transactions on Dependable and Secure Computing*, 2025.  [GitHub](#)
- Conference Paper **Mohammad Ryiad Al-Eiadeh** and Mustafa Abdallah. SAG-ERC: Securing Attack-Graph-based Systems through Node Embedding, Ranking, and Clustering. *IEEE Conference on Secure and Trustworthy CyberInfrastructure for IoT and Microelectronics (SaTC)*, Ohio, Dayton, USA, 2025.  [GitHub](#)
- Journal Article **Mohammad Ryiad Al-Eiadeh** and Mustafa Abdallah. CBDRA-IS: Centrality-Based Defense Resource Allocation for Securing Interdependent Systems. *ACM Trans. Priv. Secur.*, 2025.  [GitHub](#)
- Journal Article Al-Batah, M. S., **Mohammad Ryiad Al-Eiadeh**, & Alnsour, Y. A Novel Approach for Enhancing Plant Leaf Classification With the Binary Cuckoo Search Algorithm. *Applied Computational Intelligence and Soft Computing*, 2025.  [GitHub](#)
- Journal Article **Mohammad Ryiad Al-Eiadeh** and Mustafa Abdallah. PR-DRA: PageRank-based Defense Resource Allocation Methods for Securing Interdependent Systems Modeled by Attack Graphs. *International Journal of Information Security*, 2025.  [GitHub](#)
- Conference Paper **Mohammad Ryiad Al-Eiadeh** and Mustafa Abdallah. AARA-PR: Asset-Aware PageRank-Based Security Resource Allocation Method for Attack Graphs. *IEEE Interdisciplinary Conference on Electrics and Computer (INTCEC)*, IEEE, Chicago, IL, USA, 2024.  [GitHub](#)
- Journal Article **Mohammad Ryiad Al-Eiadeh** and Mustafa Abdallah. Genigraph: A Genetic-Based Novel Security Defense Resource Allocation Method for Interdependent Systems Modeled by Attack Graphs. *Computers & Security*, 2024.  [GitHub](#)
- Journal Article Al-Batah, M. S., & **Mohammad Ryiad Al-Eiadeh**. An Improved Binary Crow-JAYA Optimization System With Various Evolution Operators Such as Mutation for Finding the Max Clique in Dense Graphs. *International Journal of Computing Science and Mathematics*, 2024.  [GitHub](#)
- Journal Article **Mohammad Ryiad Al-Eiadeh**, Raneem Qaddoura, and Mustafa Abdallah. Investigating the Performance of a Novel Modified Binary Black Hole Optimization Algorithm for Enhancing Feature Selection. *Applied Sciences*, 2024.  [GitHub](#)

Journal Article	Al-Batah, M. S., & Mohammad Ryiad Al-Eiadeh. An Improved Discrete Jaya Optimisation Algorithm With Mutation Operator and Opposition-Based Learning to Solve the 0-1 Knapsack Problem. <i>International Journal of Mathematics in Operational Research</i> , 2023.  GitHub
Journal Article	Mohammad Ryiad Al-Eiadeh. Automatic Lung Field Segmentation Using Robust Deep Learning Criteria. <i>International Journal of Hybrid Information Technologies</i> , 2021.

– POSTER PRESENTATIONS

2026	CERIAS Annual Security Symposium, Purdue University Mohammad Al-Eiadeh and Mustafa Abdallah. <i>CBDRA-IS: Centrality-Based Defense Resource Allocation for Securing Interdependent Systems</i> .  Poster	Poster Presentation
2025	CERIAS Annual Security Symposium, Purdue University Mohammad Al-Eiadeh and Mustafa Abdallah. <i>PR-DRA: PageRank-based Defense Resource Allocation Methods for Securing Interdependent Systems Modeled by Attack Graphs</i> .  Poster	Poster Presentation

PROFESSIONAL SERVICE

2026	IEEE Reviewed 1 manuscript for <i>IEEE Transactions on Information Forensics and Security (TIFS)</i> .	Journal Peer Reviewer
2026	Springer Nature Reviewed 1 manuscript for <i>Journal of King Saud University Computer and Information Sciences</i> .  Certificate	Journal Peer Reviewer
2026	Springer Nature Reviewed 1 manuscript for <i>International Journal of Machine Learning and Cybernetics</i> .  Certificate	Journal Peer Reviewer
2025	Springer Nature Reviewed 1 manuscript for <i>Cluster Computing</i> .  Certificate	Journal Peer Reviewer
2025	Springer Nature Reviewed 1 manuscript for <i>Applied Network Science</i> .  Certificate	Journal Peer Reviewer

REFERENCE

Scholar	Purdue University, Indianapolis & West Lafayette, IN, USA Prof. Mustafa Abdallah Email: abdalla0@purdue.edu	Supervisor
Scholar	Purdue University, Indianapolis & West Lafayette, IN, USA Prof. Brian King Email: king360@purdue.edu	Co-Supervisor
Scholar	Purdue University, Indianapolis & West Lafayette, IN, USA Prof. Lingxi Li Email: lingxili@purdue.edu	Committee member
Scholar	Purdue University, Indianapolis & West Lafayette, IN, USA Prof. Maher Rizkalla Email: mrizkall@purdue.edu	Committee member
Instructor	Purdue University, Indianapolis & West Lafayette, IN, USA Sherrie Tucker Email: tucker76@purdue.edu	Assistant & Cordinator

PROJECTS

Software Project	GraphEmbedding4j: In-progress Java library for graph / node embedding research, with modular components for random walks, context-window extraction, negative sampling, skip-gram learning, gradient descent optimization, and sigmoid-based loss modeling.  GitHub
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LANGUAGES

Arabic - Native, English - Fluent